



Contact

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Nashik, Maharashtra



Education

- Bachelor of Technology
Robotics and Automation Engineering
K.K.Wagh Institute of Engineering
Education and Research, Nashik,
Maharashtra
2023 - 2027
- HSC Boards
Guru Gobind Singh Public School &
Junior College, Nashik
2021- 2023



Languages

- English - Intermediate
- Hindi - Fluent
- Marathi - Native



Technical Skills

- Programming:** Python, C, MATLAB
- Robotics & Simulation:** ROS2, URDF, RViz, Gazebo, Robot Kinematics (FK/IK, DH Parameters, Jacobians), SLAM & Navigation
- Computer Vision:** OpenCV, Image Processing, HSV Segmentation, Object & Color Detection
- Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Tkinter, Argparse, Gradio
- Tools & Platforms:** Git, GitHub, GitLab, Linux, Arduino IDE, Docker

Chaitanya Belekar

Final Year B.Tech Student
Robotics & Automation Engineering



About Me

Robotics and Automation Engineering undergraduate with hands-on experience across ROS2, computer vision (OpenCV), robot kinematics, embedded systems (Arduino), and Python application development. Skilled in modeling and simulating robotic systems using URDF, RViz, and Gazebo, with practical exposure to image processing, sensor integration, and automation workflows. Seeking an internship in robotics software, ROS2 development, computer vision, or industrial automation to deliver real-world contributions to AMR and automation systems.

PROJECT'S

Vision-Based Color Sorting Robotic Arm

2026

- Engineered a vision-guided robotic arm performing real-time object detection and color-based sorting using OpenCV with HSV segmentation, contour detection, and morphological filtering for noise reduction.
- Integrated Arduino-controlled servo motors for automated pick-and-place sorting; led hardware integration and full system calibration.

OpenKinematics Library

2026

- Developed an open-source Python library implementing forward kinematics (DH parameters) and an inverse kinematics solver for serial robotic manipulators.
- Implemented Jacobian-based velocity analysis and trajectory generation for robotic arm motion planning.
- Created interactive 3D visualizations of robot arm configurations and motion paths using Matplotlib and NumPy.

ROS2 Mobile Robot Simulation

2025

- Designed a differential-drive mobile robot model in URDF and visualized it in RViz for structural and kinematic verification.
- Developed ROS2 nodes with topic-based communication to simulate robot motion and basic sensor data exchange, including foundational concepts for AMR navigation.

Smart Password Manager (GUI & CLI)

2025

- Built a secure password manager in Python using Fernet encryption, with parallel Tkinter GUI and Argparse CLI interfaces.
- Implemented password generation and strength-analysis features following modular, secure coding practices.

Nutrition Coach ChatBot

2024

- Developed and deployed a Gradio-based conversational chatbot on Hugging Face Spaces to deliver nutrition guidance.
- Designed conversational logic to provide personalized nutrition awareness content to end users.

CERTIFICATES

Code Foundation for ROS

17th Sep, 2024

- Skills: Linux, Python3 and ROS

Python for Data Science

8th Feb, 2024

- Skills: Numpy and Pandas

MATLAB Onramp

10th Sep, 2023

- Skills: Data manipulation and visualization

LEADERSHIP & EXTRA-CURRICULAR

Event Head, RoboRace Competition — Infinity Fest, Robotics & Automation Engineering Dept.

- Led planning and execution of the department's flagship robotics racing competition, coordinating rules design, participant registration, and on-day event operations.

Plogging Activity — Nashik Ploggers

Feb 2024

- Volunteered in a community environmental cleanup drive, contributing to the collection of approximately 50 kg of waste from the Indira Nagar Jogging Track